Editors' Suggestion

Twisted chiral superconductivity in photodoped frustrated Mott insulators

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Recent advances in ultrafast pump-probe spectroscopy provide access to hidden phases of correlated matter, including light-induced superconducting states. The theoretical understanding of these nonequilibrium phases remains limited, particularly for correlated materials on frustrated lattices. Here we demonstrate that photodoping can induce a new type of chiral superconducting phase in frustrated Mott insulators by forming a condensate of doublons and holons. This metastable phase features a spatially varying order parameter with a 120° phase twist which breaks both time-reversal and inversion symmetry. Under an external electric pulse, the 120° chiral superconducting state can exhibit a second-order supercurrent perpendicular to the field in addition to a first-order parallel response, similar to a nonlinear anomalous Hall effect. Light-induced artificial gauge fields may be used to further stabilize this hidden phase. The presented results demonstrate that the light-induced superconducting state on a triangular lattice is of chiral nature and exhibits distinguishing properties which can be revealed in pump-probe experiments. The general mechanism applies to Mott insulators on various frustrated lattices and is tunable by optical means.

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I. INTRODUCTION

The last decades have witnessed an intensive search for superconducting states with spontaneously broken time-reversal and inversion symmetry. Historically the first proposal involved chemically doped Mott insulating states of high-T_c copper oxides, although the predicted properties have not been detected in experiments [1]. Since then the tantalizing chiral superconducting state, an essential building block for topological quantum computers, has been sought after in various systems, such as Sr_2RuO_4 [2], UPt_3 [3,4], and lately also in twisted bilayer graphene [5] and copper oxides [6].

In the meantime, advances in cold-atom experiments and ultrafast spectroscopy have enabled the exploration of metastable hidden states of correlated systems, which are not accessible via a thermal pathway [7–10]. A remarkable phenomenon is the putative light-induced superconductivity, which has been observed in the vicinity of Mott insulating phases [11–14]. Although the phenomena have been consistently observed in various materials, the superconducting nature and the underlying mechanisms are still under debate. One promising route to such hidden phases is photodoping, where the laser excitation drastically redistributes charges by simultaneously creating long-lived particle-like (doublon) and hole-like (holon) charge carriers [15–19]. Upon photodoping, the photocarriers may form a BEC-like superconducting condensate [13] supported by the doublon-holon exchange mediated by virtual charge recombination processes [20–23]. On bipartite lattices this may be further stabilized by a mechanism analogous to Nagaoka ferromagnetism [24,25]. A doublon-holon condensate of this type was argued to underlie the light-induced superconducting transition recently observed in a frustrated material, the triangular lattice κ-(BEDT-TTF)$_2$Cu[N(CN)$_2$]Br [13]. However, despite some supporting evidence, an unambiguous experimental signature of the doublon-holon condensate is still lacking.

As is known from quantum magnetism, frustration is intrinsically related to chiral orders [26–28]. It is thus tempting to ask whether a chiral superconducting state can be realized as a hidden phase in photodoped Mott insulators. However, the doublon-holon superconducting order has so far been discussed exclusively as a nonchiral order on bipartite or partially frustrated lattices [22,29,30]. The optical response of the nonchiral condensate largely resembles that of a conventional s-wave superconductor [22,31], therefore making it difficult to extract conclusive evidence of the pairing mechanism from pump-probe experiments.

Here we use state-of-the art numerical simulations to demonstrate that a chiral superconducting state can be stabilized by photodoping Mott insulators on fully frustrated lattices through the formation of a doublon-holon condensate with a 120° phase twist, a scenario that has been overlooked in previous studies on photocarrier condensation. Remarkably, the 120° condensate features a spatially varying order parameter and persistent currents even in the absence of an external magnetic field. Those currents change handedness under inversion or time reversal and thus define a chiral superconducting phase. In fact, the nonequilibrium nature of photodoped quasiparticles endows them with an exchange term of positive sign, an important ingredient favoring chiral condensates, especially on frustrated lattices [26]. A 120° order has also been discussed in noncollinear antiferromagnets and Floquet-driven bosonic systems [32]. Here instead we consider a condensate based on fermionic constituents, exhibiting an exotic supercurrent response that requires little or no Floquet engineering.

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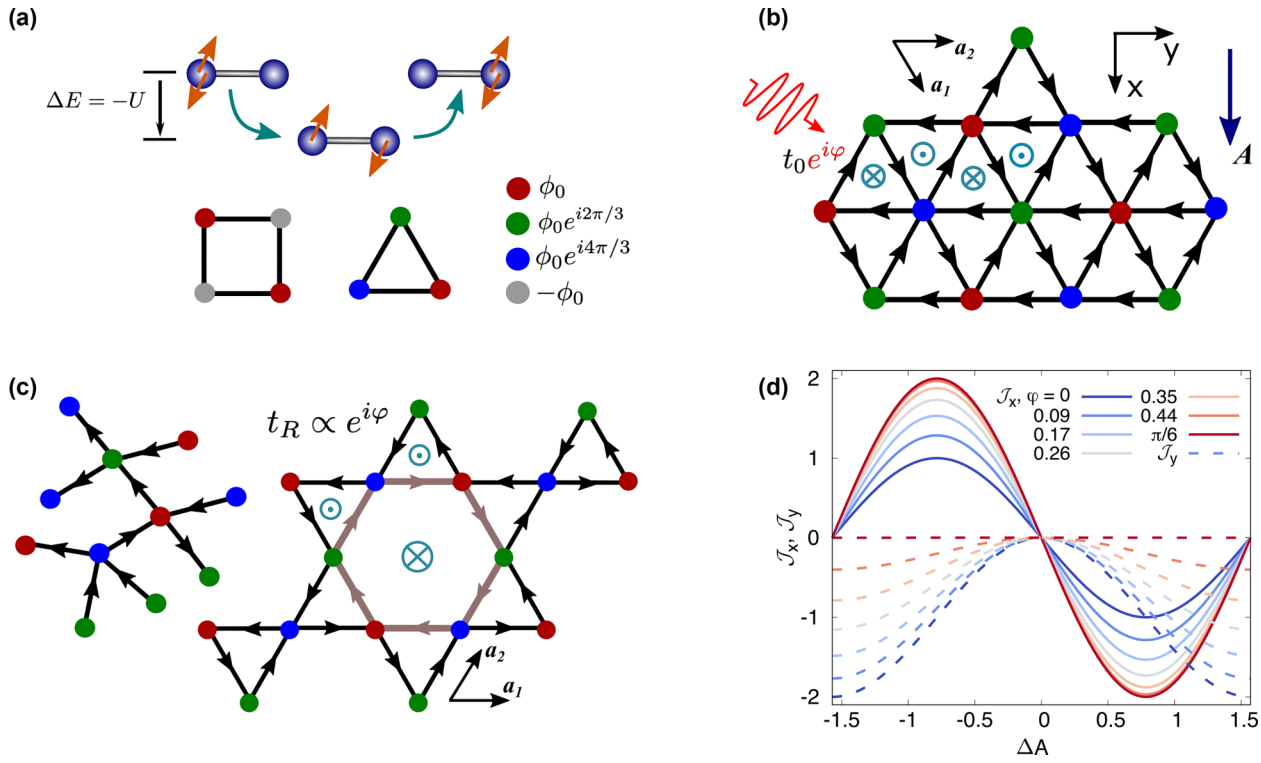


FIG. 1. Doublon-holon condensate on a three-colorable frustrated lattice. (a) The doublon-holon exchange interaction mediated by a virtual recombination. The amplitude $J_\perp$ is positive, in contrast to analogous processes in the equilibrium attractive Hubbard model. It can induce staggered η pairing on bipartite lattices and 120° pairing on three-colorable lattices. (b) The condensate on the triangular lattice. $\boldsymbol{a}_{1,2}$ are the basis vectors of the lattice. The persistent currents are indicated by black arrows. A high-frequency driving (red wavy line) induces an artificial gauge field coupling to the hopping t_0 through the Peierls phase φ, emulating an effective flux (indicated by a circle with a dot or cross) for the triangular and hexagonal motifs. This panel also shows a spatially uniform vector potential $\boldsymbol{A}$ along the x direction. (c) The condensates on the Bethe (left) and kagome (right) lattices. For the Bethe lattice (a local portion of a random regular graph frustrated at large scale) the shown 3-coloring is assumed to be the only one that is unfrustrated by the structure of large loops. For the kagome lattice one of several possible 120° condensates (the $\boldsymbol{q} = 0$ order) is shown. The magnitude of the flux through the hexagon is twice that of a triangle and of opposite sign. (d) The longitudinal ($\mathcal{J}_x$; solid lines) and transverse ($\mathcal{J}_y$; dashed lines) superconducting current density on the triangular lattice as a response to an electric pulse that shifts the vector potential by $\Delta\boldsymbol{A}$ along the x direction [with $\Delta A = 2|\Delta\boldsymbol{A}|\cos(\pi/6)$]; see panel (b). The Peierls phase φ is given in radians.

We confirm the existence of this ordered hidden phase on a Bethe lattice with infinite coordination number using nonequilibrium dynamical mean-field theory, and on the triangular and kagome lattices using exact diagonalization. The condensate is found to prevail in an extended parameter regime. We further demonstrate that an artificial gauge field created by periodic driving can be used to tune the stability and physical properties of the chiral phase [33–35]. The exotic order may possibly be realized in fermionic cold-atom systems and kagome-lattice correlated materials, such as herbertsmithite [36], and tuned by Floquet engineering. It may also be relevant for the light-induced superconducting state observed in κ-(BEDT-TTF)$_2$Cu[N(CN)$_2$]Br [13].

II. THEORETICAL ANALYSIS

A. Doublon-holon condensate on three-colorable lattices

We consider a half-filled fermionic Hubbard model under external driving, focusing on various frustrated lattices, especially the triangular and the kagome lattice. These lattices share the property of being 3-colorable (we use the color labels R, G, B) and consist of connected RGB chains or loops, i.e., triangular motifs. In order to make these lattices amenable to a dynamical mean-field study, we add a third lattice with these same properties, namely a suitable large-scale frustrated random regular graph of fixed connectivity. Any local portion of this graph looks like a Bethe lattice, see Fig. 1, which is bipartite and would thus favor η-pairing order. However, we may choose to select graphs with a large-scale structure (large loops) that frustrates the η pairing by the presence of many loops of odd length, while the particular 3-coloring shown in Fig. 1 is left unfrustrated by construction. This is taken into account by disallowing bipartite ordering, and by looking for self-consistent solutions where all sites with the same color (R, G, or B) are treated as equivalent. This mean-field solution is expected to yield the exact solution for this class of graphs in the thermodynamic limit, since the length of typical loops diverges with the size of the graph.

For any such lattice the Hamiltonian reads

$$H = -t_0 \sum_{\langle ij\rangle\sigma} (e^{i\varphi_{ij}} c^\dagger_{i\sigma} c_{j\sigma} + \text{H.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow} + g H_{\text{dr}}, \quad (1)$$

where $c_{i\sigma}$ is the electron annihilation operator at site i with spin σ, $\langle ij\rangle$ denotes nearest-neighbor pairs, and t_0 and $U > 0$ are hopping and interaction parameters. The Peierls phases φ_{ij} represent an artificial gauge field implemented through a periodic (Floquet) modulation [33,34]. The implementation of the artificial Peierls phases will be discussed at the end of the paper. As will be demonstrated below, the gauge field can be used to tune the superconducting (SC) condensate's stability, but the twisted condensate also exists for $\varphi = 0$ on the triangular and Bethe lattices. To realize a long-lived photodoped state, we assume that a driving term $gH_{\rm dr}$ with an overall amplitude g generates a nonthermal population of doublons and holons. One example of $H_{\rm dr}$ is a resonant optical excitation between the Hubbard bands, as widely adopted in experiments. Another example is the coupling of the system to two separate fermion baths (electrodes), a setup which is widely used in theoretical studies to emulate the photoexcitation protocol.

We will focus on the strong interaction regime with a weak driving $g \ll t_0 \ll U$ which is nearly resonant with the Mott gap. In this regime one generically finds a stationary nonequilibrium state whose properties are independent of the details of the driving. The system is Mott insulating in equilibrium, and the photodoped carriers have a long lifetime due to the large Mott gap [37–39]. An effective description of the photodoped state can be obtained from a $1/U$ expansion. As $g \ll t_0$, the driving term does not affect $H_{\rm eff}$ to leading order, but it can control the doublon density. Analogous to the doped Mott insulators in equilibrium, the effective physics of the photodoped state is well described by a generalized t-J model [22,23,40] $H_{\rm eff} = H_t + H_J + H_{dh}$ with hopping $H_t = -t_0 \sum_{\langle ij\rangle\sigma} e^{i\varphi_{ij}}[n_{i\bar\sigma} c^\dagger_{i\sigma} c_{j\sigma} n_{j\bar\sigma} + \bar n_{i\bar\sigma} c^\dagger_{i\sigma} c_{j\sigma} \bar n_{j\bar\sigma}] + \text{H.c.}$ and spin exchange $H_J = \sum_{\langle ij\rangle} J_{\rm ex} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$, where $\boldsymbol{S}^\alpha_i = \sum_{\sigma,\sigma'} c^\dagger_{i\sigma'} \sigma^\alpha_{\sigma'\sigma} c_{i\sigma}$ and $J_{\rm ex} = 4t_0^2/U$. We have defined $\bar n_{i\bar\sigma} = 1 - n_{i,-\sigma}$. The doublon-holon interaction term reads

$$H_{dh} = \frac{J_\perp}{2} \sum_{\langle ij\rangle} (e^{2i\varphi_{ij}} \phi_i^+ \phi_j^- + \text{H.c.}) + J_z \sum_{\langle ij\rangle} \phi_i^z \phi_j^z, \qquad (2)$$

where the pairing operators $\phi_i^+ = (\phi_i^-)^\dagger = c^\dagger_{i\uparrow} c^\dagger_{i\downarrow}$ and $\phi_i^z = (n_i - 1)/2$ span a pseudospin $\mathfrak{su}(2)$ algebra similar to that of the spin $\boldsymbol{S}_i$. The first term in Eq. (2) originates from a doublon-holon exchange process, illustrated in Fig. 1(a), which favors a doublon-holon condensation with $\langle \phi_i^+ \rangle \neq 0$. The second J_z term is a doublon attraction [the original model (1) yields $J_\perp = -J_z = J_{\rm ex}$] which by itself would induce macroscopic charge segregation. However, in reality the nearest-neighbor Coulomb repulsion will add to J_z and typically induce a moderate effective repulsion. For the maximally doped limit of $n_d = 1/2$, which maps to an $S = 1/2$ spin model, it is well known that moderate repulsion ($-J_\perp/2 < J_z^{\rm eff} < J_\perp$) leads to a homogeneous XY-ordered phase. Only once the effective doublon-doublon repulsion becomes stronger than the doublon-holon exchange, the system would develop charge density wave order. We do not consider this limit here, but anticipate phases with uniform charge density, $\langle n_i \rangle = 1$ ($\langle \phi_i^z \rangle = 0$). Note that even if the Coulomb repulsion were too weak to prevent eventual charge segregation, the ordered phase is nonetheless created by an ultrafast uniform excitation starting from a charge uniform state. We can thus assume that the state remains homogeneous on the femtosecond timescale, even if it would start to segregate on longer timescales.

In pump-probe experiments on Mott insulating solids, a strong pump pulse is often applied for a short duration to create a quasistationary photodoped state, which can also be described by the above Hamiltonian $H_{\rm eff}$ with $g = 0$. In either setup, the effective theory for the nonequilibrium state of the Hubbard model is a generalized t-J model of n_d doublons and holons and $n_s = 1 - 2n_d$ unpaired electrons per site.

In the prethermal regime, that is, before the recombination of doublons and holons becomes relevant, the positive exchange amplitude $J_\perp$ favors an opposite phase of the doublon-holon condensate on neighboring sites (η pairing) [41]. However, this alternating SC order is generically impossible on a frustrated non-bipartite lattice. Instead, we consider a 120° twisted pairing for the three-colorable lattices, defined by

$$\langle \phi^+_{i\in R} \rangle = e^{-i2\pi/3} \langle \phi^+_{i\in G} \rangle = e^{-i4\pi/3} \langle \phi^+_{i\in B} \rangle = \phi_0, \qquad (3)$$

which spontaneously breaks time-reversal and inversion symmetry. To understand the energetics of the condensate, we can examine the mean-field energy for the order $\langle \phi_i^+ \rangle = \phi_0 e^{i\boldsymbol{q}\cdot\boldsymbol{r}_i}$ with momentum $\boldsymbol{q}$, given by $\langle H_{dh}\rangle/N_{\rm site} = |\phi_0|^2 \epsilon(\boldsymbol{q})$ per site. In the following, we restrict ourselves to the case $\varphi_{ij} = \varphi$ along each bond of an $R \to G \to B$ cycle.

B. Chiral condensate on the triangular and kagome lattices

On the triangular lattice, the energy dispersion $\epsilon(\boldsymbol{q})$ reads

$$\epsilon(\boldsymbol{q}) = J_\perp[\cos(q_1 + 2\varphi) + \cos(q_2 - 2\varphi) + \cos(q_2 - q_1 + 2\varphi)]. \qquad (4)$$

The $q_{1,2}$ are the coordinates of the momentum in reciprocal space, defined as $\boldsymbol{q} = q_1\boldsymbol{b}_1 + q_2\boldsymbol{b}_2$, with reciprocal basis vectors $\boldsymbol{b}_1 = (2/\sqrt{3}, 0)$, $\boldsymbol{b}_2 = (-1/\sqrt{3}, 1)$, where we have set the lattice constant $a = 1$ to unity. The energy dispersion $\epsilon(\boldsymbol{q})$ has two minima at $\boldsymbol{q} = \pm[(-2\pi/3)\boldsymbol{b}_1 + 2\pi/3\boldsymbol{b}_2)]$ for $\varphi = 0$, corresponding to two different chiral 120° condensates. As φ becomes nonzero (and as long as $|\varphi| < \pi/3$), one of these two chiral states becomes the unique minimum in the energy landscape. The situation for $\varphi > 0$ is illustrated in the left panels of Fig. 2, where the minimum at $(2\pi/3)\boldsymbol{b}_1 + (-2\pi/3)\boldsymbol{b}_2$ is stabilized.

In the case of the kagome lattice, the energy dispersion splits into three bands because of the three atoms per unit cell. A flat band emerges as the lowest-lying one at $\varphi = 0$, and thus no particular order is singled out as energetically most favorable, as shown in the right panels. In the absence of Peierls phases, the actual ground state will be selected by quantum fluctuations, which is a complex problem not considered here. Instead we restrict our discussion of the kagome lattice to the case where we impose a pattern of φ_{ij} as shown in Fig. 1(c), which stabilizes the 120°-twisted $\boldsymbol{q} = 0$ order.

It remains to be shown that the 120° condensate is stable against quantum fluctuations. We first consider the maximum photodoping situation ($n_d = 0.5$) where all sites are either doubly occupied or empty. In this case $H_{\rm eff} = H_{dh}$ corresponds to an XXZ model of pseudospin $\boldsymbol{\phi}$. As mentioned before, for this limit it is well established that on

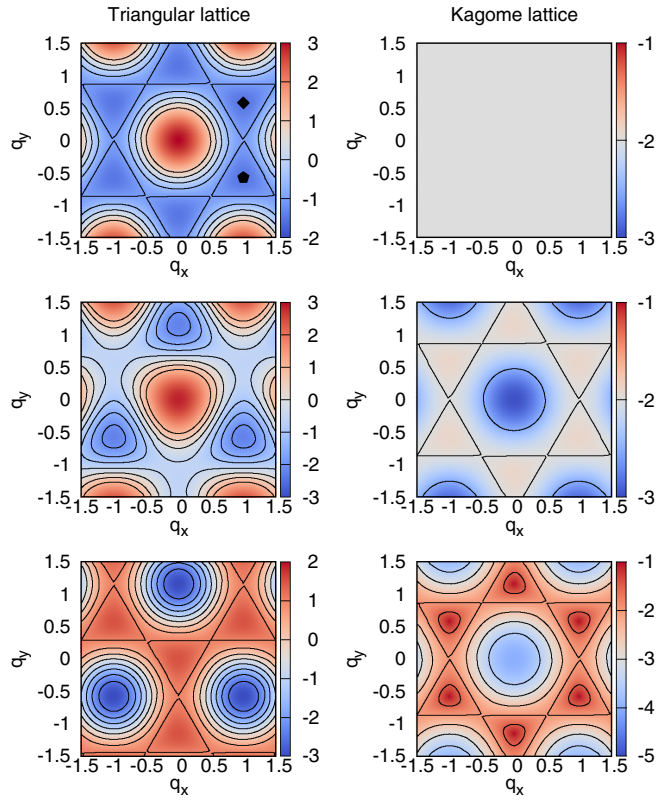


FIG. 2. The energy dispersion $\epsilon(\boldsymbol{q})$ for orders with momentum $\boldsymbol{q}$ on the triangular and the kagome lattices. Various values for the Peierls phase $\varphi = 0, \pi/20, \pi/6$ are shown in the first, second, and third row, respectively. The $q_{x,y}$ denote the Cartesian coordinates in reciprocal space, measured in units of $2\pi/\sqrt{3}$. The left panels show the energy dispersion for the triangular lattice. The top panel exhibits two minima at $\boldsymbol{q} = \pm[(2\pi/3)\boldsymbol{b}_1 + (-2\pi/3)\boldsymbol{b}_2]$, which are degenerate for $\varphi = 0$ and marked by a black pentagon and diamond, respectively. One minimum, $\boldsymbol{q} = (2\pi/3)\boldsymbol{b}_1 + (-2\pi/3)\boldsymbol{b}_2$ (pentagon), is selectively stabilized for $0 < \varphi < \pi/3$. For $\varphi = \pi/6$ the stabilization is optimal, and the dispersion becomes nearly isotropic around the minimum. This entails the absence of leading-order trigonal warping and the vanishing of the second-order supercurrent response. The right panels show the lowest-lying band of the kagome lattice. For $\varphi = 0$ the band is flat, while $\varphi > 0$ stabilizes the $\boldsymbol{q} = 0$ condensate.

the triangular lattice the 120° phase is realized for effective doublon-doublon interactions in the window $J_z/J_\perp \in [-1/2, 1]$. For the specific case that the effective repulsion takes the value $J_z = J_\perp \cos(2\pi/3 + 2\varphi)$ ($J_z = -J_\perp/2$ for $\varphi = 0$), the two possible 120° condensates actually span the exact ground-state manifold for both the triangular and kagome lattices considered here; see Appendix B for a proof following the idea of Ref. [42]. Away from maximum photodoping, the twisted 120° condensate is weakened by the presence of singly occupied sites and the additional terms in H_{eff}, such as electron hopping. However, the condensate should survive at least for $1/2 - n_d \ll O(t_0/U)$. In Sec. III, we will numerically confirm that this condensate is in fact stable in an extended parameter regime away from the maximum photodoping limit $n_d = 1/2$.

C. Anomalous optical response of the 120° chiral condensate on the triangular lattice

The doublon-holon condensate generally coexists with unpaired electrons, resulting in a gapless state with a large metallic conductivity and negative AC conductivity, which reflects the possibility of stimulated recombination [31]. Meanwhile, the twisted 120° condensate embodies a spatially varying phase twist and thus carries a persistent superconducting current even in the absence of an external field. Indeed, the phase-twisted condensate can be viewed as a frustrated array of Josephson junctions [43], and a persistent current $\mathcal{J}^{dh} = 2\mathcal{J}_0^{dh}\sin(2\pi/3 + 2\varphi)$ flows along the $R \to G \to B$ direction, due to the phase twist $2\pi/3$; see arrows in Fig. 1. Each triangular motif (either upward- or downward-pointing triangle) is circled by a current of order $eJ_\perp \sim et_0^2/U$, giving rise to a magnetic moment of the order of $\mu \sim eJ_\perp a^2/\hbar \sim \mu_B(t_0/U)(m_e/m_{\text{eff}})$, where μ_B is the Bohr magneton and $m_{\text{eff}} = \hbar^2/a^2 t_0$ is an effective mass related to the electronic bandwidth. For a lattice constant $a = 0.1$ nm and $J_\perp = 0.1$ eV, e.g., one has $\mu \sim 10^{-25}$ A m$^2 \approx 0.01\mu_B$. This weak magnetic moment should not strongly affect the superconducting condensate. The net current cancels out on macroscopic length scales when the loop currents on the three bonds have identical magnitude.

A macroscopic superconducting current arises, however, once a shift of the vector potential is induced via an electric pulse, $\Delta\boldsymbol{A} = -\int_{t_i}^{t_f} \boldsymbol{E}dt$. Neglecting potential phase slips, the phase difference across a link ij increases by $A_{ij} = \Delta\boldsymbol{A}\cdot(\boldsymbol{r}_i - \boldsymbol{r}_j)$, which affects the currents on the three types of bonds differently. We assume $\varphi_{ij} = \varphi$ for the Peierls phases on a $B \to G \to R$ chain. For a bond $\langle ij\rangle$, the doublon-holon contribution to the current, evaluated at the mean-field level, is $\mathcal{J}_{ij}^{dh}(A_{ij}) = \delta H_{dh}/\delta A_{ij} \approx 2\mathcal{J}_0^{dh}\sin(2\pi/3 + 2\varphi + 2eA_{ij})$, where $\mathcal{J}_0^{dh} = -2eJ_\perp|\phi_0|^2$ with the elementary charge e.

Let us consider the topmost triangular motif shown in Fig. 1(b). Writing $\Delta\boldsymbol{A} = (\Delta A_x, \Delta A_y)$, we obtain for A_{ij} on the three different bonds

$$\begin{aligned} A_{RG} &= \frac{1}{2}\Delta A_y - \frac{\sqrt{3}}{2}\Delta A_x, \\ A_{GB} &= \frac{1}{2}\Delta A_y + \frac{\sqrt{3}}{2}\Delta A_x, \\ A_{BR} &= -\Delta A_y, \end{aligned} \tag{5}$$

where the lattice constant was set to unity. The bond currents are accordingly given by $\mathcal{J}_{ij} = \mathcal{J}^{dh}(A_{ij})$.

We can express the macroscopic (average) current density as

$$\mathcal{J}^a = D^{ab}\Delta A_b + T^{abc}\Delta A_b \Delta A_c \tag{6}$$

for $a, b, c = x/y$, where the coefficient tensors D and T are constrained by symmetry. First, the presence of a rotational C_3 symmetry implies $D^{ab} \propto \delta^{ab}$. Moreover, the 120° condensate is invariant under reflection with respect to the y axis ($x \to -x$), which entails that $T^{xyy} = T^{xxx} = T^{yxy} = 0$. The invariance of T under C_3 rotation finally imposes $-T^{yyy} = T^{yxx} = T^{xxy}$. Note that the mixed components of T imply that to second order in ΔA the current has a component transverse to the applied electric field.

Let us now determine the macroscopic current response of the triangular lattice. The gauge-invariant current density is generally obtained as $\boldsymbol{J} = \delta H/\delta \boldsymbol{A}$. On the triangular lattice, assuming the absence of phase slips the application of a current pulse simply shifts the wave vector of the condensate, $\boldsymbol{q}_0 \to \boldsymbol{q}_0 + \Delta\boldsymbol{A}$. On the mean-field level, the expectation value of the Hamiltonian per unit cell can be written as

$$\frac{\langle H\rangle}{N_{\text{site}}S_{\text{uc}}} = \frac{\epsilon(\boldsymbol{q}_0+\Delta\boldsymbol{A})}{S_{\text{uc}}} = \frac{J_\perp\phi_0^2}{S_{\text{uc}}}\sum_\alpha \cos(2\pi/3+2\varphi+2\Delta A_\alpha), \quad (7)$$

where α runs over three bonds, RG, GB, BR, $\boldsymbol{q}_0 = (2\pi/3, -2\pi/3)$, and the area of the unit cell is $S_{\text{uc}} = \sqrt{3}/2$. The stationarity of the quasiequilibrium state with wave vector $\boldsymbol{q}_0$ implies a vanishing current density $\propto \partial\epsilon/\partial\boldsymbol{q}|_{\boldsymbol{q}=\boldsymbol{q}_0}$ in the global minimum. The linear supercurrent response is then determined by the Hessian $D^{ab} \propto \partial^2\epsilon/\partial\boldsymbol{q}_a\partial\boldsymbol{q}_b|_{\boldsymbol{q}=\boldsymbol{q}_0}$. The second-order response tensor is given by $T^{abc} \propto \partial^3\epsilon/\partial\boldsymbol{q}_a\partial\boldsymbol{q}_b\partial\boldsymbol{q}_c|_{\boldsymbol{q}=\boldsymbol{q}_0}$, which is nonzero due to the finite trigonal warping. Explicitly, we find the current density

$$\mathcal{J}_x = \frac{\partial\epsilon(\Delta\boldsymbol{A})}{\partial A_x}\frac{1}{S_{\text{uc}}} = \mathcal{J}_{GB} - \mathcal{J}_{RG}, \quad (8)$$

$$\mathcal{J}_y = \frac{\partial\epsilon(\Delta\boldsymbol{A})}{\partial A_y}\frac{1}{S_{\text{uc}}} = (\mathcal{J}_{RG} + \mathcal{J}_{GB} - 2\mathcal{J}_{BR})/\sqrt{3}. \quad (9)$$

Expanding to second order in $\Delta\boldsymbol{A}$, one finds the components of the D and T tensors as

$$D^{xx} = D^{yy} = 2\sqrt{3}J_0^{dh}\cos(2\pi/3+2\varphi), \quad (10)$$

$$T^{yxx} = -T^{yyy} = T^{xxy} = -\frac{\sqrt{3}}{2}J_0^{dh}\sin(2\pi/3+2\varphi). \quad (11)$$

As we noted above, the T tensor vanishes for $\varphi = \pi/6$, since the leading term of the trigonal warping vanishes at that point. This can also be read off from Fig. 2, where the energy dispersion near the minimum becomes nearly isotropic, which suppresses the third-order derivative. This and the overall dependence of the current response on the artificial gauge field φ is plotted in Fig. 1(d).

The nonlinear and transverse current response is phenomenologically similar to a second-order anomalous Hall effect in the presence of inversion-symmetry breaking, as discussed in Refs. [44,45]. However, in contrast to the latter effect, the present transverse current appears in the context of a superconducting response and requires explicitly the breaking of time-reversal symmetry. It is interesting to note that if one applies a continuous-wave sinusoidal driving along x, $E_x(t) = E_0\sin(\omega t)$, the transverse current J_y oscillates at frequency 2ω. Such a second harmonic transverse response could serve as a distinctive probe of the chiral order that is accessible via a standard four-point measurement.

III. NUMERICAL DETERMINATION OF THE PHASE DIAGRAM

A. Methods

In the remainder of the article, we will explore the phase diagram of the chiral condensate on three different frustrated lattices: the triangular and kagome lattices and a constrained random regular graph where a specific tricoloring is singled out by its large-scale structure. The solution of the latter is obtained by solving a Bethe lattice problem with a fixed 3-coloring of sites, using nonequilibrium dynamical mean-field theory. The triangular and kagome lattices will be studied via exact diagonalization on finite-size clusters. In this subsection, we briefly describe the two methods and their implementation.

1. Nonequilibrium dynamical mean-field theory

We solve the driven Hubbard model on the Bethe lattice with infinite coordination number using nonequilibrium dynamical mean-field theory (DMFT) [46]. The model can be exactly mapped to an Anderson impurity model when the hopping is rescaled as $t_0/\sqrt{z}$ for $z \to \infty$, which leads to a noninteracting bandwidth $4t_0$. The exchange interaction for each bond is given by $J_{\text{ex}} = 4t_0^2/zU$, while its total (mean-field) energy contribution scales as z times this value, $4t_0^2/U$. This infinitely coordinated system captures local correlation effects and, in particular, the interplay between hopping and exchange interactions, while it completely neglects nonlocal correlations as well as the effects of electrons hopping around loops, present in lower-dimensional systems.

The lattice problem is exactly mapped to three single-impurity Anderson models on the Keldysh contour, defined by the action (spin index suppressed for simplicity of notation)

$$S_{X,\text{imp}} = \int dt\, c_X^\dagger(t)(i\partial_t - h_{\text{loc}})c_X(t) - \int dt\,dt'\, c_X^\dagger(t)\left[\Delta_X(t-t') + \sum_\ell D_\ell(t-t')\right]c_X(t'), \quad (12)$$

where $X = R, G, B$ and $\Delta_X(t,t')$ is the hybridization function representing the lattice environment. $D_{\ell=\pm}$ represent the hybridization functions of auxiliary baths used for the photodoping. For an R site, half of its neighbors are G sites and the other half are B, and similarly for the other two sites. The local Hamiltonian h_{loc} includes the Hubbard interaction and the pair seed term. In practice, we apply a seed term $h\sum_i \theta_i c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger + \text{H.c.}$ with $h = 0.001$ to break the symmetry, where $\theta_i = 1, e^{i2\pi/3}, e^{i4\pi/3}$ for the R, G, B site, respectively.

The three impurity problems labeled by R, G, B are solved with the noncrossing approximation [47,48], which yields the corresponding impurity Green's functions $G_{\text{imp}}(t,t')$ and the doublon number (per site) $n_d = \langle n_\uparrow n_\downarrow\rangle$. The new hybridization function Δ is determined from the local Green's function $G_{\text{loc}}(t,t') = G_{\text{imp}}(t,t')$ by the Bethe-lattice self-consistency [49]

$$\Delta_R(t,t') = t_0^2\tau_z(e^{i\tau_z\varphi}G_B(t,t')e^{-i\tau_z\varphi} + e^{-i\tau_z\varphi}G_G(t,t')e^{i\tau_z\varphi})\tau_z/2, \quad (13)$$

and analogously for $\Delta_{G/B}$, maintaining a bandwidth of $4t_0$. φ is the Peierls phase and the Pauli matrix τ_z appears because of the Nambu formalism [22].

Within DMFT, the current flowing through a site R (average of the currents flowing out to the neighboring B sites and

flowing in from the neighboring G sites) can be calculated as

$$J = -\tfrac{1}{2}\,\mathrm{Re}\,\tau_x G_R(t,t') * \big[t_0^2\tau_z(e^{i\tau_z\varphi}G_B(t,t')e^{-i\tau_z\varphi} - e^{-i\tau_z\varphi}G_G(t,t')e^{i\tau_z\varphi})\tau_z/2\big], \quad (14)$$

and similarly for the other sites. Here, the symbol $*$ represents a convolution on the Keldysh contour.

a. Steady-state formalism. In the nonequilibrium steady-state setup, the driving term reads $gH_{\mathrm{dr}} = \frac{g}{\sqrt{L}}\sum_{i\sigma\alpha}(c_{i\sigma}^\dagger d_{i\alpha\sigma} + \mathrm{H.c.}) + \sum_{i\alpha\sigma}\epsilon_\alpha d_{i\alpha\sigma}^\dagger d_{i\alpha\sigma}$ with bath operators $d_{i\alpha\sigma}$ and $\alpha = (\ell,\xi)$ containing a bath label $\ell = \pm$ and the energy level label ξ. L represents the bath size and has the dimension of a length. In practice, we use two fermion baths with a semielliptic density of states with temperature T_b and chemical potentials $\mu_b^\pm$. The two baths are designed to create doublons by injecting electrons from the upper bath (with $\mu_b^+ \sim U/2$) into the upper Hubbard band and to create holons by transferring electrons from the lower Hubbard band to the lower bath (with $\mu_b^- \sim -U/2$). More details can be found in Ref. [22]. The baths can be exactly integrated out and incorporated into the DMFT iterations through the hybridization density of states $D_\pm(\epsilon) = \Gamma\sqrt{1-(\epsilon \pm U/2)^2/W^2}$ with $\Gamma = g^2/W$ and half-bandwidth W. The parameters $\mu_b^\pm$ and T_b are varied to control the doublon and holon distribution in the Hubbard system. The steady-state problem is solved with a frequency-domain strong-coupling impurity solver, whose implementation is detailed in Ref. [50].

b. Real-time formalism. The real-time dynamics of the system is simulated using an entropy-cooling protocol [51,52]. To be concrete, the Hubbard system is coupled to two narrow bands at each site, which have semielliptic densities of states of half-bandwidth 0.1 centered at the energies $\omega_\pm = \pm 6$. The upper band (ω_+) is empty while the lower band (ω_-) is full, as the chemical potential is set to $\mu_b = 0$. We drive the coupling constant $g = v(t)$ with a pulse of frequency $\Omega(t)$, inducing a resonant charge transfer between the empty (full) narrow band and the lower (upper) Hubbard band, respectively. The key idea is to adjust the frequency in time, so as to match the first Floquet sideband of the narrow band with the effective Fermi level for the doublons or holons [roughly speaking one wants $\omega_- + \Omega(t) \sim \mu^+(t)$, where μ^+ is the Fermi level in the upper Hubbard band, and similarly for the lower Hubbard band]. See Ref. [52] for more details. Specifically, the pulse is given by $v(t) = \frac{\sin[\Omega(t)t]}{(1+e^{(t-t_1-T)\gamma_{\mathrm{off}}})(1+e^{-(t-t_1)\gamma_{\mathrm{on}}})}$ for $t > 0$ and zero otherwise, where $\Omega(t) = \Omega_i + (\Omega_f - \Omega_i)\sin(\pi t/400)$ and $t_1 = 4$, $T = 100$, $\gamma_{\mathrm{on}} = 1$, $\gamma_{\mathrm{off}} = 1/4$, $\Omega_i = 7.25$, $\Omega_f = 12.5$. The frequency $\Omega(t)$ is varied to fill the upper Hubbard band to its top, and to empty the lower Hubbard band to its bottom. The parameters are empirically optimized to minimize T_{eff} in the final state and generate a long-lived order.

2. Exact diagonalization

In the exact diagonalization studies, we solve the generalized t-J model on the triangular and kagome lattices with an artificial gauge field φ. Three-site terms are ignored, since they play a similar role as the electron hopping, assisting doublon/holon delocalization, but have a much lower strength $J_{\mathrm{ex}} \ll t_0$. As we discussed earlier, a moderate short-range Coulomb repulsion should in principle be added to prevent charge segregation in the thermodynamic limit. However, on the small systems considered in this study, we neglect this term and detect the overall tendency to form a pair condensate by carrying out a site average of the relevant correlation functions.

Three conserved quantities, the number of single up and down spins and the doublon number n_d, are imposed to satisfy $n_\uparrow = n_\downarrow = 1/2 - n_d$, valid at half filling. For the triangular lattice, a fully spatially symmetric 12-site cluster is solved with the Lanczos algorithm to obtain the ground-state wave function. We define $\boldsymbol{a}_1$ ($\boldsymbol{a}_2$) as in Fig. 1(b), and the torus is spanned by $2\boldsymbol{a}_1 + 2\boldsymbol{a}_2$ and $2\boldsymbol{a}_1 - 4\boldsymbol{a}_2$. The pairing correlation is averaged over sites, using translational invariance and the structure factor defined in Sec. III C is summed over chiralities (replacing $\theta_i \to \theta_i + \theta_i^*$).

In the case of the kagome lattice, a fully spatially symmetric 12-site cluster is solved with the same method. We consider off-resonant polarized light in the long-wavelength limit. The hopping term along bond $\langle ij\rangle$ is then dressed with the Peierls phase $\exp[i\boldsymbol{r}_{ij}\cdot\boldsymbol{A}(t)] = \exp[i(\mathcal{A}_{ij}e^{i\Omega t} + \mathcal{A}_{ij}^* e^{-i\Omega t})]$, where $\boldsymbol{r}_{ij} = \boldsymbol{r}_i - \boldsymbol{r}_j$ and $\mathcal{A}_{ij} = \mathcal{A}_0 a e^{i\theta_{ij}}/2$ for a bond parallel to $(\cos\theta_{ij}, \sin\theta_{ij})$.

The lth Fourier component of the Hamiltonian reads

$$H_l = \delta_{l0}\sum_i U_i n_{i\uparrow}n_{i\downarrow} - i^{|l|}t_0\sum_{\langle ij\rangle\sigma} e^{il\theta_{ij}}J_{|l|}(\mathcal{A})c_{i\sigma}^\dagger c_{j\sigma}, \quad (15)$$

where $\mathcal{A} = \mathcal{A}_0 a/2$ and the J_l are Bessel functions of the first kind. In the high-frequency limit, one can obtain the effective Hamiltonian with a $1/\Omega$ expansion,

$$H_0 + \sum_{l>0}[H_l, H_{-l}]/l\Omega = \sum_i U_i n_{i\uparrow}n_{i\downarrow} - t_{\mathrm{R}}\sum_{\langle ij\rangle\sigma} c_{i\sigma}^\dagger c_{j\sigma} - t_{\mathrm{NNN}}\sum_{\langle\langle ik\rangle\rangle\sigma} c_{i\sigma}^\dagger c_{k\sigma}. \quad (16)$$

The complex nearest-neighbor (NN) hopping and purely imaginary next-NN hopping read

$$t_{\mathrm{R}} = t_0 J_0(\mathcal{A}) - it_0^2\sum_l (-)^l J_l(\mathcal{A})^2\sin(2l\pi/3)/l\Omega,$$
$$t_{\mathrm{NNN}} = -it_0^2\sum_l (-)^l J_l(\mathcal{A})^2\sin(l\pi/3)/l\Omega. \quad (17)$$

We include both hoppings in the exact diagonalization.

B. Bethe lattice

To provide a concrete example of the 120° twisted order, we first consider a numerically solvable model, the driven Hubbard model on the Bethe lattice with hopping $t_0/\sqrt{z}$, where the coordination number is taken to $z \to \infty$ and t_0 sets the unit of energy. The solution of this model in the strong-interaction regime can be obtained using nonequilibrium DMFT [46,53]; see Sec. III A 1. As explained above we impose a given 3-coloring on the Bethe lattice (with competing orders suppressed by large-scale frustration).

We first discuss the results obtained with the nonequilibrium steady-state setup [50] described in Sec. III A 1 a. Figure 3(a) shows the spectral function $A = -\frac{1}{\pi}\,\mathrm{Im}\,G^r$ and

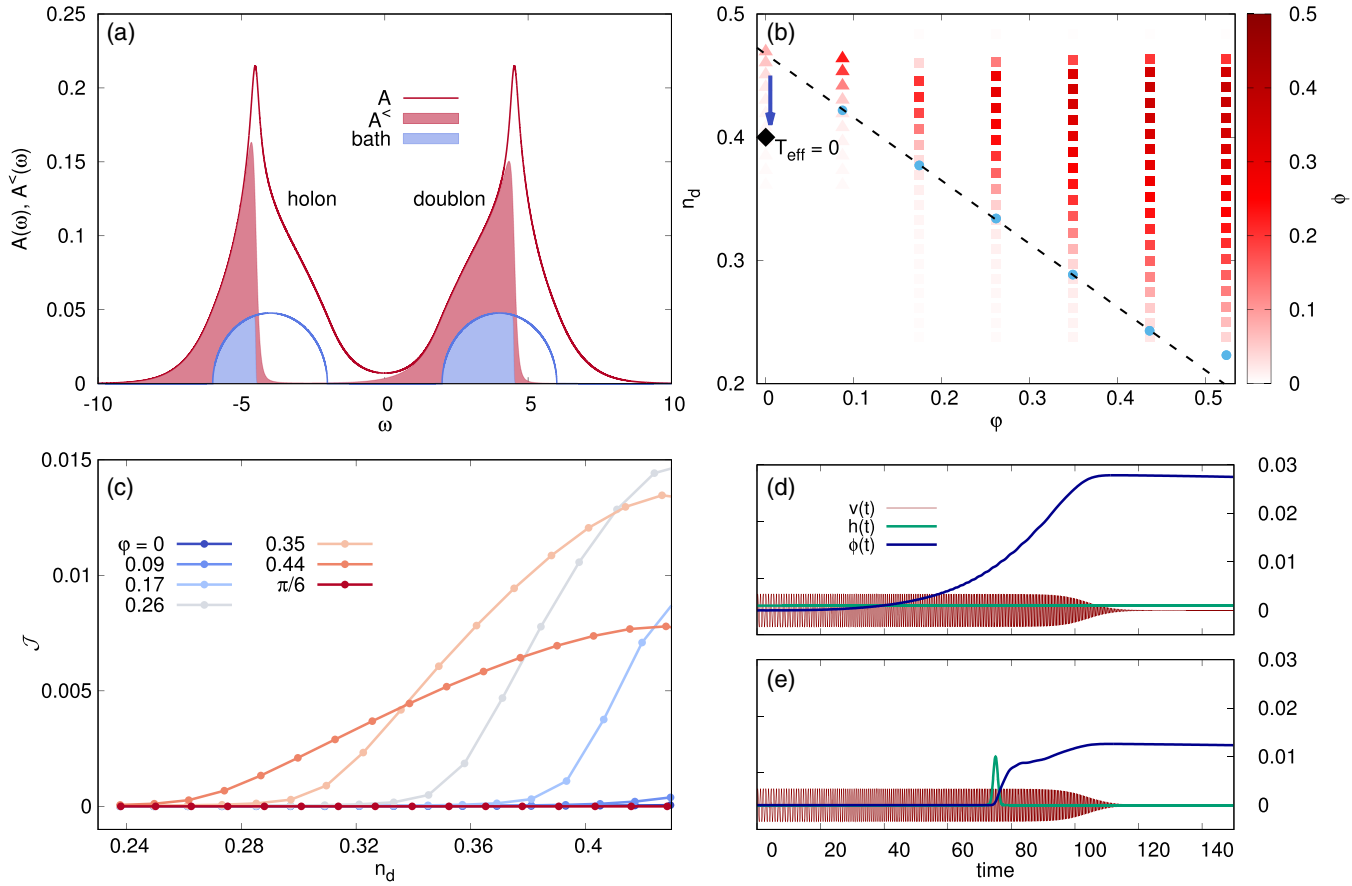


FIG. 3. The chiral 120° condensate on the Bethe lattice. In panels (a)–(c) the system is driven by two fermion baths, where the "upper" bath injects high-energy electrons to form doublons, while the "lower" bath absorbs low-energy electrons from singly occupied sites to form holons (steady-state DMFT setup). Panels (d) and (e) show real-time simulations with an explicit photodoping pulse. (a) Spectral function and occupation at $\varphi = 0$ for the driven Bethe lattice, with $\mu_b^{\pm} = \pm 4.5$, $T_b = 0.01$, $\Gamma = 0.045$, $U = 8.0$, and bandwidth 4 (in units of t_0). The red-shaded area indicates the occupation of the Hubbard bands, while the blue-shaded area indicates the bath occupation. (b) The n_d-φ phase diagram for the 120° condensate sampled by varying the parameters of the fermion baths. The red color indicates the magnitude of the order parameter $|\phi|$. The phase φ is measured in radians and varies from 0 to $\pi/6$. $W = 2$ for the square symbols and $W = 2.7$ for the triangles. The blue dots are obtained by fitting the phase boundary. The black diamond is the critical n_d for $\varphi = 0$, extrapolated to $T_{\rm eff} = 0$. (c) The persistent current evaluated with $T_b = 0.01$, $W = 2$. A weak symmetry-breaking field $h = 0.001$ is used in the DMFT iterations. (d) and (e) Transient 120°-ordered states obtained by means of the entropy-cooling protocol for the Bethe lattice with $U = 9.0$ and $\varphi = 0$ using two ways to break the symmetry by a small pairing field [green curves $h(t)$]. $v(t)$ is the oscillating system-bath coupling which produces the photodoped state. It is plotted with arbitrary units.

the occupation $A^< = \frac{1}{2\pi} \mathrm{Im}\, G^<$ for a typical driven state. The external driving creates two separate Fermi levels in the lower and upper Hubbard bands around $\omega = \mu_b^{\pm}$, indicating the presence of excess doublons and holons. The two Fermi energies correspond to the energy cost or gain associated with the replacement of a doublon or holon by an unpaired electron. Even though the system is highly excited, the Fermi edges can be rather sharp, with low effective temperatures $T_{\rm eff}$ defined by fitting the distribution function $f(\omega) = A^<(\omega)/A(\omega)$ separately near the two Fermi levels. This confirms that the system can be described by a constrained quasiequilibrium state with excess doublons and holons of density n_d and temperature $T_{\rm eff}$, as already shown in previous works [22,23,50]. The order parameter $\phi = \langle \phi^+ \rangle$ is sampled for various bath parameters, which allows us to generate the phase diagram for the 120° condensate shown in Fig. 3(b). The system exhibits a transition to the 120°-ordered phase beyond a critical double occupancy n_d^c, indicated by blue dots (see Appendix D), and is enhanced as φ increases. A persistent current $\mathcal{J}$ flows along $R \to G \to B$. Its magnitude is plotted in panel (c). In experiments, carefully designed protocols can minimize the entropy production and thus achieve low effective temperatures [51]. In our calculations, we can tune the effective temperature of the doublons and holons by changing bath parameters, and then extrapolate the phase boundary to $T_{\rm eff} = 0$. This procedure yields an upper bound of the critical doublon number $n_d^c \approx 0.4$ at $\varphi = 0$ [black diamond in (b)]; see Appendix C for details.

The above steady-state DMFT results for the Bethe lattice confirm the existence of the 120° SC condensate in a wide parameter range, which extends to $\varphi = 0$ and at least down to $n_d \approx 0.4$, and also the validity of the effective theory (2) at finite U. It remains to be investigated whether this intriguing state can be realized on an ultrafast timescale. To address this question, we consider a real-time entropy-cooling protocol [51,52], which allows us to generate cold photodoped

states. The implementation of this scheme is described in Sec. III A 1 b. As shown in Fig. 3(d), a nonzero 120° SC order parameter quickly emerges at $\varphi = 0$ under the driving $v(t)$, in the presence of a small (but not infinitesimal) symmetry-breaking field $h = 0.001$ coupling to ϕ_i^+. In the case of transition metal compounds, our unit of time is of the order of femtoseconds. Within the time range accessible in our numerical simulations, the order appears to decay very slowly after the driving is switched off. In addition, we have also simulated a system perturbed by a short pulse of the symmetry-breaking field, $h(t)$; see Fig. 3(e). In this case, the 120° order continues to grow after the pulse, which strongly suggests a spontaneous symmetry breaking. Note that the final order parameter which is established spontaneously in Fig. 3(e) is small, which indicates that the system is close to the phase transition. In this regime, the presence of even a small constant field h leads to a significant enhancement of the order parameter, which explains the larger value in panel (d).

C. Triangular lattice

Next, we use exact diagonalization to directly treat the generalized t-J model (2) on a triangular lattice, to address the existence of the condensate in 2D systems at $T_{\text{eff}} = 0$. We calculate the pairing structure factor at the 120° point,

$$S(120) = \sum_{ij} \theta_i \theta_j^* \langle \phi_i^+ \phi_j^- \rangle / N_{\text{site}}, \tag{18}$$

with $\theta_{i\in R} = e^{i2\pi/3}\theta_{i\in G} = e^{i4\pi/3}\theta_{i\in B} = 1$ corresponding to the "color" of the site. $S(120)$ quantifies the total 120° order. The result is shown in Fig. 4(a), and suggests the stability of the 120° condensate away from $n_d = 1/2$ down to $\varphi = 0$, in qualitative agreement with the DMFT results for the Bethe lattice. The critical doublon number is $n_d^c(\varphi = 0) \sim 0.37$.

D. Kagome lattice

We have demonstrated that an artificial Peierls phase can be used to tune the stability of the twisted 120° condensate, but have not yet discussed how to implement it in practice. In cold-atom systems, a staggered flux similar to Fig. 1(b) can be readily created through a periodic shaking of the lattice, as discussed for example in Ref. [32].

It is more challenging to create the artificial Peierls phase in solid-state systems. On the triangular lattice, the twisted SC phase is stabilized already for $\varphi = 0$ and the Peierls phase is not mandatory. The situation however differs on the kagome lattice. The generalized t-J model in the limit of maximal photodoping ($n_d = 0.5$) is equivalent to an XXZ spin model, which exhibits a complex phase diagram on the kagome lattice. As discussed above, the mean-field $\epsilon(\boldsymbol{q})$ is completely flat in the absence of φ_{ij}, giving rise to a macroscopically degenerate ground-state manifold. Our key task is to induce an artificial gauge field which favors a specific 120° ordering pattern, for example the $\boldsymbol{q} = 0$ three-coloring of the lattice shown in Fig. 1(c). In the case of solid-state systems, we consider applying a strong laser with circular polarization, similar to the setup in Ref. [35], with a vector potential $\boldsymbol{A}(t) = \mathcal{A}_0(\cos(\Omega t), -\sin(\Omega t))$. The field couples through a time-dependent Peierls phase $\exp[i(\boldsymbol{r}_i - \boldsymbol{r}_j)\cdot\boldsymbol{A}(t)]$ to the bond $\langle ij\rangle$. In the high-frequency limit ($\Omega \gg t_0$, off-resonant with U), the hopping parameters are renormalized, leading to a complex NN hopping $t_R = |t_R| e^{i\varphi(\mathcal{A})}$. This renormalized hopping appears to favor the $\boldsymbol{q} = 0$ condensate shown in Fig. 1(c). The next-NN hopping which is also created and would favor a uniform order is usually very small; see Appendix D.

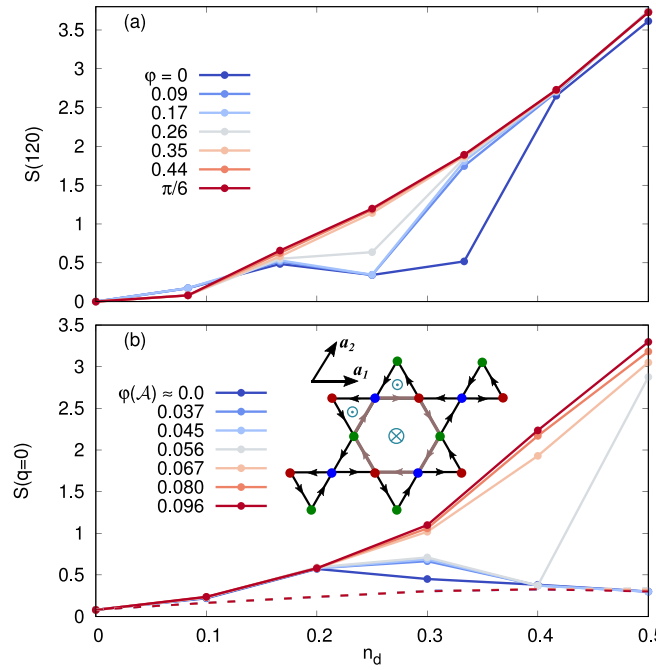


FIG. 4. Exact diagonalization results for the triangular and kagome lattices for a half-filled nonmagnetized system, $n_\uparrow = n_\downarrow = 1/2 - n_d$. (a) The pairing structure factor $S(120)$ of the 120°-twisted condensate versus the doublon number per site n_d for the 12-site triangular lattice at $T_{\text{eff}} = 0$. The Peierls phase φ is varied from 0 to $\pi/6$. (b) The pairing structure factor of the ($\boldsymbol{q} = 0$) 120° order as a function of the doublon density for the 12-site kagome lattice. The amplitude $\mathcal{A}$ of the dressing field is varied from 0 to 1.4 to produce different φ values, as indicated in the label. Here, an external driving with frequency $\Omega = 5$ is assumed. The dashed line shows the structure factor for the uniform order. $J_\perp = -J_z = 0.5$ is assumed.

We study the resulting effective model on a 12-site cluster with exact diagonalization, defining $\mathcal{A} = \mathcal{A}_0 a/2$ with lattice constant $a = 1$. The results are shown in Fig. 4(b), and clearly indicate the appearance of the ($\boldsymbol{q} = 0$) 120° SC condensate for $\varphi(\mathcal{A}) \gtrsim 0.067$ and $n_d \gtrsim 0.3$. The dependence of $S(120)$ on n_d is qualitatively similar for both the triangular and kagome lattices for n_d above their respective transitions. In contrast, the uniform order (indicated by the dashed line) is always suppressed. We note, however, that other potentially competing orders, such as a $\sqrt{3}\times\sqrt{3}$ type 120° order, cannot be studied with the small cluster shown in Fig. 1. The two orders share the identical mean-field energy at $\varphi = 0$. However, a nonzero $|\varphi| < \pi/3$ stabilizes the $\boldsymbol{q} = 0$ order considered here; see Appendix D. The $\sqrt{3}\times\sqrt{3}$ type order can, however, be stabilized with a different artificial gauge field. Finally we note that other terms, which are generated by the Floquet driving, but were ignored here, may affect the precise location of the critical values $n_d^c(\varphi)$.

IV. CONCLUSIONS

Our work establishes a new type of chiral superconductivity in photodoped Mott insulators, which breaks the time-reversal and inversion symmetries through a spatially twisted order parameter—the 120° condensate. This condensate originates from a positive doublon-holon exchange amplitude $J_\perp > 0$, which is intrinsically related to the nonequilibrium nature of the photodoped states and contrasts with the negative exchange in equilibrium BEC-like pairing induced by charge attraction [54]. The exchange processes furthermore generate a doublon-holon interaction (the J_z term), which favors charge segregation but in realistic solids is counteracted by repulsive Coulomb interactions. In the presence of a light-induced artificial gauge field, the chiral order can be further enhanced and be stabilized on the kagome lattice as well. The persistent loop current and the nonlinear transverse superconducting current are characteristic signatures of the chiral 120° condensate and allow us to realize a second harmonic generation as well as possible photovoltaic effects due to the rectification property of the second-order response [45]. This phenomenology contrasts with the conventional description of the SC electromagnetic response based on the linear London equation $\boldsymbol{j} \propto \boldsymbol{A}$, where even-order responses are excluded by time-reversal or inversion symmetry.

The optical conductivity of the doublon-holon condensate is expected to be similar to the bipartite case [22], which exhibits a normal nonsuperconducting Drude peak, together with a supercurrent response and a negative conductivity at $\omega \sim U$ that reflects the carrier recombination at sufficiently high photodoping. In the presence of a single domain of the chiral condensate, an AC field will induce a transverse superconducting current that could be observed with a four-point measurement.

In experiments, photodoped states can be realized by applying femtosecond laser pulses to condensed matter systems [8], or via the tilting of optical lattices [55]. Either protocol allows us to create long-lived photodoped doublons and holons in the presence of a large Mott gap [37–39]. Long before those can recombine, they relax to a prethermal quasiequilibrium governed by a generalized t-J model. Our work shows that if the doublon-holon excitation is carried out while maintaining a low entropy, and if a significant number of doublon excitations is achieved, the prethermal state will relax to a chiral condensate.

Since on the triangular lattice the condensate exists in the absence of artificial gauge fields ($\varphi = 0$), it may be relevant to the photodoped SC states in correlated materials living on triangular lattices, such as κ-(BEDT-TTF)$_2$Cu[N(CN)$_2$]Br [13]. However, for this specific material, the hopping amplitude t' along one bond (e.g., along $\boldsymbol{a}_2$-$\boldsymbol{a}_1$) is different from the hopping t_0 along the other two bonds. By evaluating the partially frustrated energy dispersion (see Appendix A), we find that chiral SC order is still favored in the regime $t' \gtrsim 0.7t_0$, whereas a staggered η-paired SC state takes over when $t' < 0.7t_0$. Different estimates for the ratio t'/t_0 in κ-(BEDT-TTF)$_2$Cu[N(CN)$_2$]Br have been reported, ranging from about 0.24 [13] to about 1.0 [56,57].

It is also worth noting that our analysis suggests that the chiral condensate is most stable against thermal fluctuations when the material is on the insulating side of the Mott transition, but close to it. Indeed, there $J_{\rm ex} \propto 1/U$ is maximized, while a photodoped state remains well defined. However, the pump energy in the mentioned experiments may well have been too low to produce a sufficient number of photocarriers to induce a chiral phase. It would be interesting to pump the κ salts with higher frequencies directly across the Mott gap, especially in insulating materials, such as κ-(BEDT-TTF)$_2$Cu[N(CN)$_2$]Cl. Finally, doped Mott insulators have a very rich phase diagram, and the competition between the chiral condensate and other long-lived and hidden phases, especially away from maximum photodoping, is an interesting topic for further investigations.

ACKNOWLEDGMENTS

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APPENDIX A: FRUSTRATION CROSSOVER

In this Appendix we study the crossover from the square lattice to the triangular lattice by continuously tuning the hopping parameter of one bond. To be specific, we assume that the "strong" bonds along $\boldsymbol{a}_1, \boldsymbol{a}_2$ still have the hopping amplitude t_0, while the hopping along the "weak" bond $\boldsymbol{a}_2$-$\boldsymbol{a}_1$ has the amplitude $\sqrt{\alpha}t_0$ where α is tuned from 0 (square lattice) to 1 (triangular lattice). This setup is relevant to the scenario of κ-(BEDT-TTF)$_2$Cu[N(CN)$_2$]Br, where different estimates have suggested $\sqrt{\alpha}$ values from ∼0.24 to ∼1 [13].

When $\alpha = 0$, the lattice is bipartite and a staggered η-paired order is expected to be favored, while $\alpha = 1$ corresponds to the triangular lattice where the chiral 120° order is favored. We are interested in the crossover between the two orders.

For general α, the energy dispersion becomes

$$\epsilon(\boldsymbol{q})/J_\perp = \cos(q_1) + \cos(q_2) + \alpha\cos(q_2 - q_1) = f(\boldsymbol{q}). \tag{A1}$$

We now search for the stationary points of $f(\boldsymbol{q})$. Requiring $\partial_1 f = \partial_2 f = 0$ gives $\sin q_1 = -\sin q_2$, leading to $q_1 = -q_2 + 2k\pi$ or $q_1 = q_2 + (2k+1)\pi$ for $k \in \mathbb{Z}$.

We first consider the stationary points with $q_1 = (2k+1)\pi + q_2$. In this case $\partial f/\partial q_1 = -\sin q_1 = \partial f/\partial q_2 = -\sin q_2 = 0$. In the first Brillouin zone $(-\pi, \pi] \times (-\pi, \pi]$, the possible solutions are $(0, \pi)$ and $(\pi, 0)$, and the energy values are both $-\alpha$. The stability of the stationary points are controlled by the eigenvalues of the Hessian matrix $\left(\begin{smallmatrix} \partial_1^2 f & \partial_{12}^2 f \\ \partial_{12}^2 f & \partial_2^2 f \end{smallmatrix}\right)$, given by $\lambda_{1,2} = \alpha \pm \sqrt{\alpha^2 + 1} > 0$, always leading to two saddle points.

We then consider the stationary points at $q_1 = -q_2 + 2k\pi = q$, where the eigenvalues of the Hesse matrix are

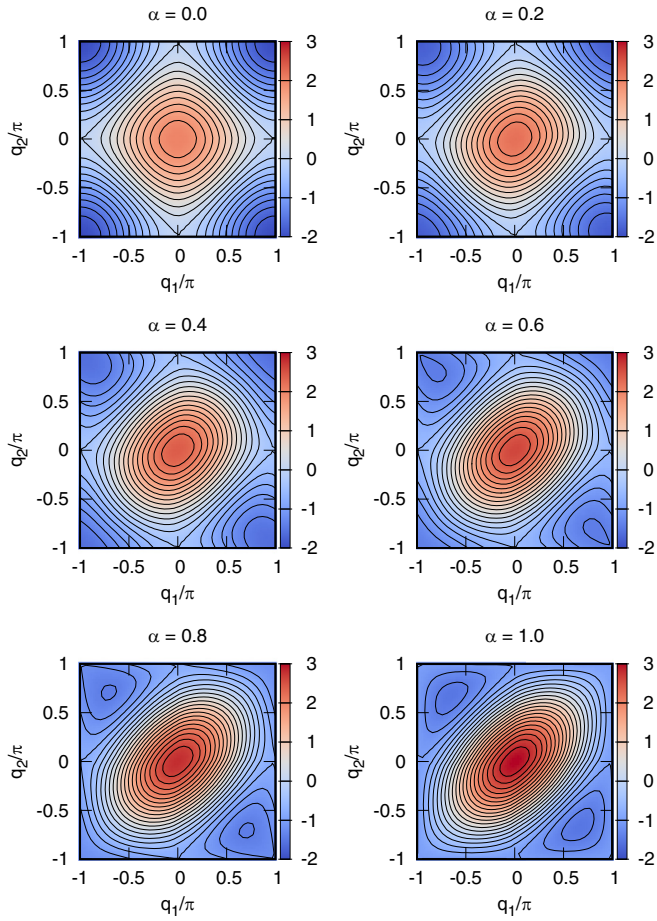


FIG. 5. Energy dispersion $\epsilon(\boldsymbol{q})$ for orders with momentum $\boldsymbol{q}$ on the triangular lattice with partial frustration α. $\varphi = 0$ is assumed. q_1 and q_2 are the reciprocal coordinates which define $\boldsymbol{q} = q_1\boldsymbol{b}_1 + q_2\boldsymbol{b}_2$. In the regime $\alpha \geqslant 0.5$, the minimum at $\boldsymbol{q} = \pi\boldsymbol{b}_1 + \pi\boldsymbol{b}_2$ (η pairing) splits into two minima which continuously cross over to the 120° order at $\boldsymbol{q} = \pm[(-2\pi/3)\boldsymbol{b}_1 + (2\pi/3)\boldsymbol{b}_2]$.

$\lambda_1 = -\cos q$ and $\lambda_2 = -\cos q - 2\alpha\cos 2q = -4\alpha\cos^2 q - \cos q + 2\alpha$.

From this, one can identify the following two regimes:

(1) When $\alpha < 0.5$, only one stationary point at $q = \pi$ exists, giving rise to a minimum at (π, π) since $\lambda_1 = 1, \lambda_2 = 1 - 2\alpha > 0$. This is the η-pairing case. In this regime, the system has one minimum (π, π) with $f = -2 + \alpha < -1.5$, and η pairing is always dominant.

(2) When $\alpha > 0.5$, the (π, π) minimum becomes a saddle point since $\lambda_2 = 1 - 2\alpha < 0$. It splits into two new minima at $\pm(q_m, -q_m)$ with $q_m = \pm\arccos(-1/2\alpha)$. This solution has $\lambda_1 = 1/2\alpha > 0, \lambda_2 = (2\alpha - 1)(1 + 1/2\alpha) > 0$. In this regime, one has two minima at $\pm(q_m, -q_m)$ with the energy value $f = -3\alpha/2$, corresponding to the two chiral orders.

This crossover behavior can be seen from Fig. 5. In summary, $\alpha = 0.5$ is the transition point where the η order becomes unstable and two chiral minima emerge. The chiral order can be generally incommensurate with the lattice periodicity for an arbitrary α, giving rise to a family of pair-density-wave states. This state evolves to the 120° order at $\alpha = 1$, the fully frustrated point. The transition point $\alpha = 0.5$ corresponds to a weak-bond hopping of $\sqrt{0.5} \approx 0.7$ times the strong-bond hoppings.

APPENDIX B: EXACTLY SOLVABLE POINT OF THE DOUBLON-HOLON EXCHANGE HAMILTONIAN

For triangular and kagome lattices, the Hamiltonian H_{eff} can be decomposed as $H_{\text{eff}} = \sum_\Delta h(\Delta)$, i.e., into a sum of triangular motifs Δ. In the following, we will prove that the 120° condensate is the exact ground state for $J_z = J_\perp\cos(2\pi/3 + \varphi)$ and $0 < \varphi < \pi/3$, following the argument of Changlani *et al.* [42]. For simplicity, we use pseudospin $|\uparrow\rangle$ and $|\downarrow\rangle$ to denote doublon and holon states, respectively.

More precisely, we will prove that the coherent state of hard-core doublons

$$|\Psi\rangle = \exp\left(\sum_i \theta_i^* \phi_i^+\right)|0\rangle \tag{B1}$$

is a ground state of the Hamiltonian if $J_z = J_\perp\cos(2\pi/3 + \varphi)$. Here the sum is over all sites i, and the "color" θ_i is defined as in the main text. This state is essentially a tensor-product state $\bigotimes_i |C_i\rangle$ with $C_i = R, G, B$ at each site i, defined as $|R\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$, $|G\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + \omega|\downarrow\rangle)$, $|B\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + \omega^2|\downarrow\rangle)$ with $\omega = e^{i2\pi/3}$.

The Hilbert space of each motif is spanned by eight orthogonal states, including the fully polarized $|3/2\rangle = |\uparrow\uparrow\uparrow\rangle$, $|-3/2\rangle = |\downarrow\downarrow\downarrow\rangle$ and two sets of chiral states defined by

$$\begin{aligned}
|+1/2\rangle_L &= |\downarrow\uparrow\uparrow\rangle + \omega|\uparrow\downarrow\uparrow\rangle + \omega^2|\uparrow\uparrow\downarrow\rangle,\\
|-1/2\rangle_L &= |\uparrow\downarrow\downarrow\rangle + \omega^2|\downarrow\uparrow\downarrow\rangle + \omega|\downarrow\downarrow\uparrow\rangle,\\
|+1/2\rangle_R &= |\downarrow\uparrow\uparrow\rangle + \omega^2|\uparrow\downarrow\uparrow\rangle + \omega|\uparrow\uparrow\downarrow\rangle,\\
|-1/2\rangle_R &= |\uparrow\downarrow\downarrow\rangle + \omega|\downarrow\uparrow\downarrow\rangle + \omega^2|\downarrow\downarrow\uparrow\rangle,
\end{aligned} \tag{B2}$$

and two nonchiral states

$$\begin{aligned}
|+1/2\rangle_0 &= \frac{1}{\sqrt{3}}(|\downarrow\uparrow\uparrow\rangle + |\uparrow\downarrow\uparrow\rangle + |\uparrow\uparrow\downarrow\rangle),\\
|-1/2\rangle_0 &= \frac{1}{\sqrt{3}}(|\uparrow\downarrow\downarrow\rangle + |\downarrow\uparrow\downarrow\rangle + |\downarrow\downarrow\uparrow\rangle).
\end{aligned} \tag{B3}$$

The above states are labeled by the total z pseudospin ϕ^z (number of doublons) and the momentum $\pm 2\pi/3, 0$ associated with the threefold rotational symmetry. It is crucial to note that $|\pm\rangle_L$ are degenerate and $|\pm\rangle_R$ are degenerate due to the symmetry under a combined reflection and a particle-hole transformation (the exchange of $|\uparrow\rangle$ and $|\downarrow\rangle$). Indeed, the above six states are all eigenstates of the Hamiltonian with eigenvalues $\lambda_{\frac{3}{2},0} = \lambda_{-\frac{3}{2},0} = 3J_z/4$, $\lambda_{\frac{1}{2},L} = \lambda_{-\frac{1}{2},L} = -J_z/4 + J_\perp\cos(2\pi/3 + 2\varphi)$, $\lambda_{\frac{1}{2},R} = \lambda_{-\frac{1}{2},L} = -J_z/4 + J_\perp\cos(2\pi/3 - 2\varphi)$, and $\lambda_{\frac{1}{2},0} = \lambda_{-\frac{1}{2},0} = -J_z/4 + J_\perp\cos(2\varphi)$, where we have labeled the eigenvalues by $\lambda_{\phi^z_{\text{tot}},c}$ with chirality $c = R, L, 0$. We can verify this by explicit calculations. For $|\pm\rangle_L$ one has

$$\begin{aligned}
h(\Delta)|+1/2\rangle_L = {} & e^{-i2\varphi}|\uparrow\downarrow\uparrow\rangle + e^{i2\varphi}|\uparrow\uparrow\downarrow\rangle + e^{-i2\varphi}\omega|\uparrow\uparrow\downarrow\rangle\\
& + e^{i2\varphi}\omega|\downarrow\uparrow\uparrow\rangle + e^{-i2\varphi}\omega^2|\downarrow\uparrow\uparrow\rangle + e^{i2\varphi}\omega^2|\uparrow\downarrow\uparrow\rangle\\
= {} & \lambda_{\frac{1}{2},L}|+1/2\rangle_L,
\end{aligned}$$

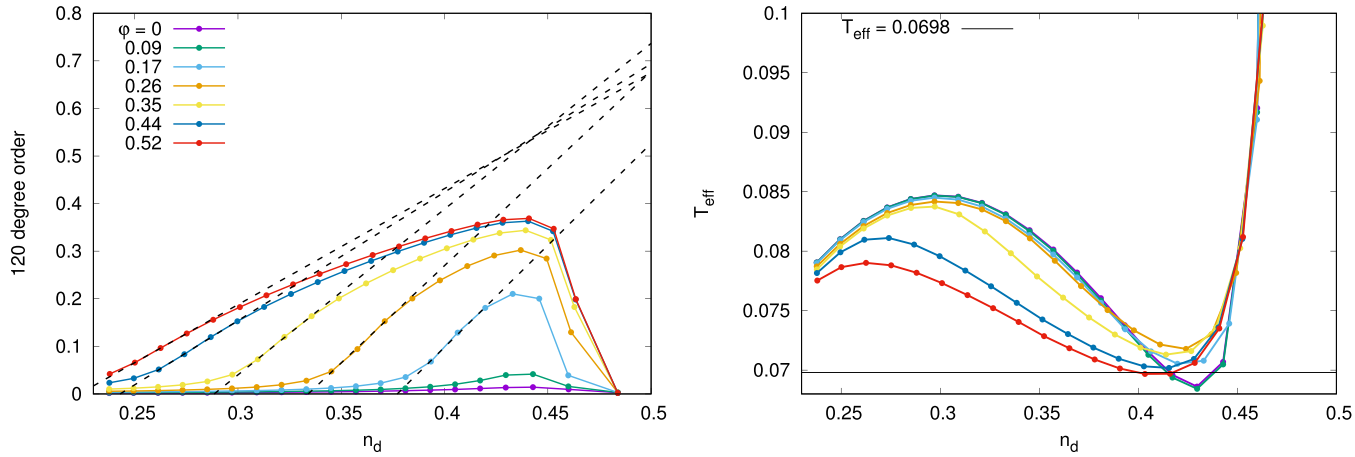


FIG. 6. Determination of the phase boundary by fitting the order parameter (left panel) and the effective temperature for $T_b = 0.01$, $W = 2$ (right panel). The effective temperature is estimated as the minimum of the curves with $\varphi > 0.09$.

$$h(\Delta)|-1/2\rangle_L = e^{i2\varphi}|\downarrow\uparrow\downarrow\rangle + e^{-i2\varphi}|\downarrow\downarrow\uparrow\rangle + e^{i2\varphi}\omega^2|\downarrow\downarrow\uparrow\rangle + e^{-i2\varphi}\omega^2|\uparrow\downarrow\downarrow\rangle + e^{i2\varphi}\omega|\uparrow\downarrow\downarrow\rangle + e^{-i2\varphi}\omega|\downarrow\uparrow\downarrow\rangle = \lambda_{-\frac{1}{2},L}|-1/2\rangle_L, \tag{B4}$$

which are similar for the chiral pairs. The single-motif Hamiltonian $h(\Delta)$ can then be recast into the form

$$H = \sum_{\Delta} h(\Delta) = \lambda_{\frac{1}{2},0}\sum_{\Delta}\mathcal{P}_0(\Delta) + \lambda_{\frac{1}{2},R}\sum_{\Delta}\mathcal{P}_R(\Delta) + \lambda_{\frac{1}{2},L}\sum_{\Delta}\mathcal{P}_L(\Delta) + \lambda_{\frac{3}{2}}\sum_{\Delta}\mathcal{P}_{3/2}(\Delta), \tag{B5}$$

where $\mathcal{P}_{0,R,L,3/2}$ projects the states to the nonchiral, R, L, and fully polarized subspaces for a triangle Δ, respectively. When $J_z = J_\perp\cos(2\pi/3 + 2\varphi)$, the fully polarized states and one set of chiral states $|+1/2\rangle_L$ and $|-1/2\rangle_L$ are degenerate as $\lambda_{\frac{1}{2},L} = \lambda_{\frac{3}{2}}$. For $0 < \varphi < \pi/3$, the states of the opposite chirality and the nonchiral states lie higher in the energy spectrum of $h(\Delta)$.

A key observation is that, for a single triangular motif, the three-coloring state reads

$$|R\rangle|G\rangle|B\rangle = \frac{1}{2^{3/2}}[|+3/2\rangle + |-3/2\rangle + (|\uparrow\downarrow\downarrow\rangle + \omega^2|\downarrow\uparrow\downarrow\rangle + \omega|\downarrow\downarrow\uparrow\rangle) + (|\downarrow\uparrow\uparrow\rangle + \omega|\uparrow\downarrow\uparrow\rangle + \omega^2|\uparrow\uparrow\downarrow\rangle)] = \frac{1}{2^{3/2}}[|3/2\rangle + |-3/2\rangle + |+1/2\rangle_L + |-1/2\rangle_L], \tag{B6}$$

which only contains fully polarized states and $|\pm 1/2\rangle_L$. As a result, at $J_z = J_\perp\cos(2\pi/3 + 2\varphi)$, the coherent state $|\Psi\rangle$ consistently zeros out $\mathcal{P}_R$ and $\mathcal{P}_0$, and must be a ground state with energy $3J_zN_\Delta/4$, where N_Δ is the number of triangular motifs.

This three-coloring state is an exact ground state for all ϕ^z sectors, thus equally favoring the 120° condensate and charge segregation (uniform ϕ^z). For the triangular lattice, when $|J_z| < |J_\perp\cos(2\pi/3 + 2\varphi)|$ one can argue that the 120° order is favored. This parameter regime is probably most realistic in experiments due to the intersite Coulomb repulsion. We note that a relatively strong nearest-neighbor Coulomb repulsion $V\sum_{\langle ij\rangle} n_in_j$ can change J_z to $J_z = V - 4t_0^2/(U - V) > 0$, favoring charge-density-wave order, which is frustrated on the lattices considered here.

The stability against deviations from maximum photodoping ($n_d = 1/2$) has been numerically studied in the main text. At $\varphi = \pi/6$, since the order commutes with the electron-hopping operator, one expects an extended range of stable 120° condensate, at least when the energy scale of the hopping n_st_0 is not much stronger than the exchange energy $J_{\rm ex} \sim 4t_0^2/U$, namely $n_s = 1 - 2n_d \lesssim 4t_0/U$.

APPENDIX C: PHASE BOUNDARY AND THE CRITICAL DOUBLON NUMBER FOR THE BETHE LATTICE

In this Appendix, we discuss the fitting of the phase boundary and the effective temperature. The left panel of Fig. 6 plots the order parameter as a function of n_d, which is varied by changing the bath chemical potentials $\mu_b^\pm = \pm\mu_b$. The data are fitted by dashed lines near the phase transition, and the critical doublon densities n_d^c are estimated as the vanishing points of the dashed lines. The transition is not sharply defined because of the weak symmetry-breaking field $h = 0.001$ introduced in the DMFT iterations. In Fig. 3(b) the data points below the fitted phase boundary have been cut off. For $W = 2$, the effective temperature in the steady-state setup dramatically shoots up for $n_d \gtrsim 0.43$, resulting in a strongly suppressed order parameter, and the phase boundary cannot be precisely determined in the above way. In this regime, both the external fermion baths and the upper/lower Hubbard bands of the system become close to full or empty, leading to a bottleneck for the thermalization and energy dissipation. This bottleneck can be partially overcome by using $W = 2.7$. However, the intraband thermalization is generically reduced for a nearly full/empty band.

We have also varied T_b to obtain stationary states with different $T_{\rm eff}$, yielding a function $n_d^c(\varphi, T_{\rm eff})$. $T_{\rm eff}$ generally

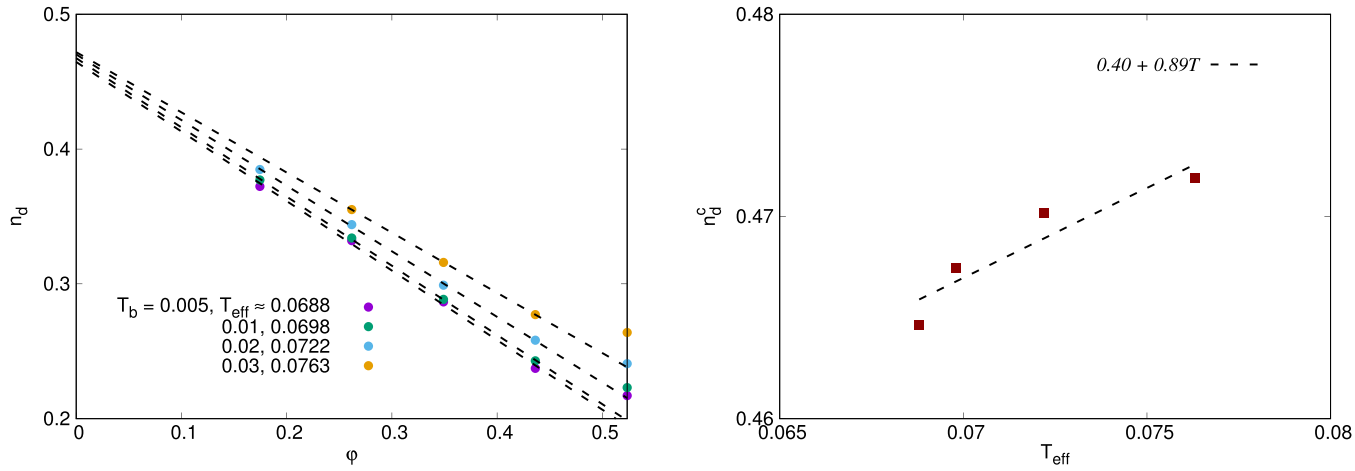


FIG. 7. Fitting of the phase boundary and temperature effects. The data are obtained by varying T_b and the corresponding T_{eff} is estimated as in the text. The fermion bath has $W = 2$. The left panel shows fits of the phase boundary which are extrapolated to $\varphi = 0$. The extrapolated $\varphi = 0$ critical doublon number is consistent with the order parameter obtained with the $W = 2.7$ bath. The right panel shows the extrapolated $n_d(\varphi = 0)$ (the intercept) and the slope of the dashed lines in the left panel as a function of T_{eff}. From these values we can extrapolate to obtain the rather conservative upper bound $n_d \approx 0.40$ and the phase boundary at $T_{\text{eff}} \to 0$.

varies along each curve for fixed φ, and as a conservative estimate, we have used the lowest value for $\varphi > 0.09$ for all data points (see the horizontal line in the right panel of Fig. 6). This is a crude approximation which tends to underestimate T_{eff}, and it thus guarantees that we get a *lower bound* for the order parameter at a given T_{eff}. The function $n_d^c(\varphi)$ is determined in the same way for each data set with different T_b. We have not fitted the n_d^c for $\varphi = 0$ and $\varphi = 0.09$, since the order parameters are substantially smaller in these two cases and the fit becomes less robust.

To estimate the critical doublon density at $T_{\text{eff}} = 0$ and $\varphi = 0$, we first fit the curves $n_d^c(\varphi, T_{\text{eff}})$ for each T_b (with the estimated T_{eff}) and extrapolate to obtain an estimate of $n_d^c(0, T_{\text{eff}})$; see the left panel of Fig. 7. The n_d^c data points for varying φ and $T_b = 0.01$ have been shown in Fig. 3, and we can see that the extrapolated $n_d^c(0, T_{\text{eff}})$ is consistent with the data for $\varphi = 0$ and $W = 2.7$. We then extrapolate the function $n_d^c(0, T_{\text{eff}})$ to $T_{\text{eff}} = 0$, as shown in the right panel of Fig. 7.

APPENDIX D: THE DRIVEN HUBBARD MODEL ON THE KAGOME LATTICE

Floquet driving with circularly polarized light creates a complex phase for the hopping parameter t_{R} up to the lowest order in a $1/\Omega$ expansion. In fact, it adds to the original t_0 a purely imaginary term at the order t_0^2. The resulting doublon-holon exchange term is of order $t_{\text{R}}^2/U \sim t_0^2/U + O(t_0^3/U)$. Other terms generated by the light are of higher orders, including a purely imaginary next-NN hopping at the order of t_0^2, which generates an exchange term of $O(t_0^4/U)$. Therefore, we believe the main effect is from the renormalization of t_{R}. We also note that some Floquet-generated terms, such as the chiral spin interaction proportional to $\boldsymbol{S}_i \cdot (\boldsymbol{S}_j \times \boldsymbol{S}_k)$ [35], can nevertheless alter the phase diagram of the system at the quantitative level.

In this Appendix, we compare the mean-field energy of the uniform, the $\boldsymbol{q} = 0$, and the $\sqrt{3} \times \sqrt{3}$ phases (see Fig. 8) under various driving amplitudes $\mathcal{A}$. Specifically, we assume $\phi_0 = 1/2$ and evaluate the doublon-holon interaction term with a mean-field decoupling $\langle \phi_i^+ \phi_i^- \rangle \to \phi_0^2 \theta_i \theta_j^*$, where θ_i is defined as in the pairing structure factor. It suffices to calculate and compare the mean-field energy for a single hexagon. The driving-induced NN and next-NN hoppings are shown in Fig. 9(a) and the corresponding mean-field energies are shown in Fig. 9(b). The $\boldsymbol{q} = 0$ 120° order has the lowest energy among the three orders, consistent with the ED results shown in the main text.

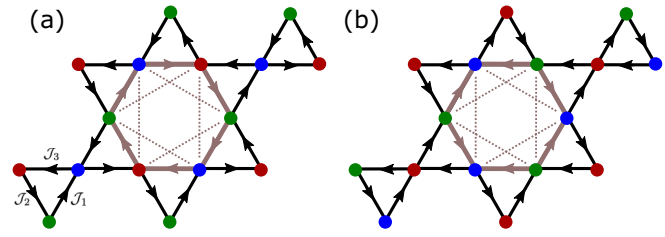


FIG. 8. Hubbard model on the kagome lattice engineered by circularly polarized laser driving. The NN hopping is renormalized. The induced (second-order in t_0) next-NN hopping is shown for one (light-brown) hexagon. (a) When the NN hopping t_{R} has a phase close to $\pi/6$, it favors the $\boldsymbol{q} = 0$ order. (b) The next-NN hopping favors identical order parameters on sites connected by the next-NN bond, possibly favoring the uniform order or the $\sqrt{3} \times \sqrt{3}$ order shown here.

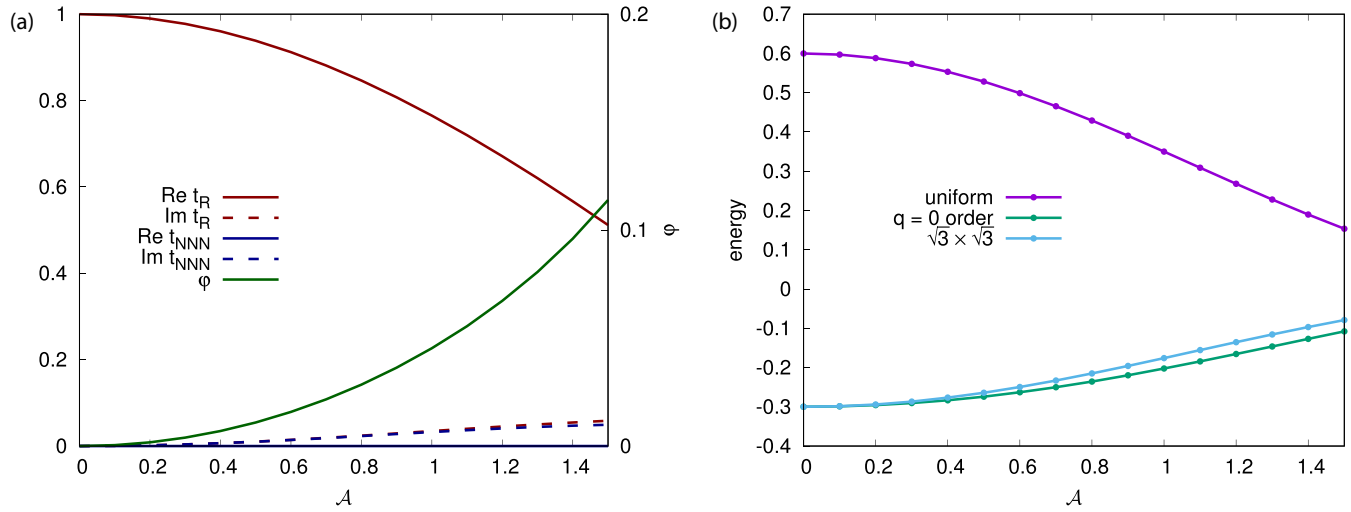


FIG. 9. The hopping parameters (a) and semiclassical energy (b) for different orders under driving, with $|\boldsymbol{\phi}| = 0.5$ assumed. The parameters are $U/t_0 = 8$, $\Omega/t_0 = 5$. The $\boldsymbol{q} = 0$ mode generally has the lower energy than the $\sqrt{3} \times \sqrt{3}$ state. In the strong-driving limit, the phase of t_R approaches $\pi/2$, and the uniform order becomes significantly enhanced.

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